

Fig. 2: Actual-size PCB layout of the sensing peripheral devices circuit

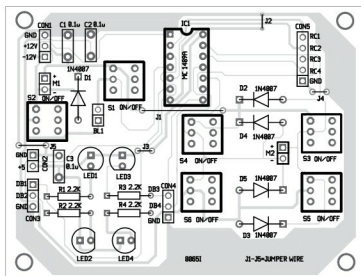


Fig. 3: Component layout of the PCB

to these receivers. You can add protection resistors, capacitors and diodes for inputs of IC1, if required. Also, you can interface any micro-controller (MCU) directly to IC1. Data outputs (DB1 through DB4) of MC1489A can be read with simple input commands from the MCU.

Switches S1 through S6 can be of any kind, be it electronic (transistors) or electromechanical (relays). Ensure that the voltages supplied to MC1489A through these switches do not exceed its operating range. Maximum input voltage range of MC1489A is $\pm 30V$.

Construction and testing

Actual-size, single-side PCB layout of the circuit for sensing and controlling peripheral devices with MC1489A

receiver is shown in Fig. 2 and its component layout in Fig. 3.

Testing the circuit is very simple. If you do not have actual loads, you can replace bulb BL1 and motors M1 and M2 with appropriate resistors. For example, you may replace these with resistors of 200-ohm, 3W for power supply of $\pm 12V$. This will provide currents of $24V/200\text{-ohm} = 120\text{mA}$ through loads, when corresponding switches are closed. This current will imitate the working current of the loads.

Switch off and switch on switches S1 through S6 in any working combination. You will notice that the

status of the LEDs changes according to the status of the switches.

Pay special attention to the controlling of DC motor M2 in the bridge configuration.

Closing switches S3 through S6 using the right combination is essential; otherwise, there will be a short circuit. Never close switches S3 and S5, and/or S4 and S6, at the same time. To turn on M2, you can close only S3 and S6, or S4 and S5, switches at the same time. ●

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